

Akhil Gurrapu

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SUMMARY

Data, AI & Application Engineer with 7 years of experience architecting and delivering robust, scalable enterprise data platforms across the modern data stack. Currently a Senior Snowflake Engineer on an enterprise Platform Team, with deep expertise administering Snowflake from the ground up — RBAC and access governance, platform infrastructure, Infrastructure as Code with Terraform, Jenkins-based CI/CD, performance tuning, and FinOps cost optimization. Experienced building end-to-end Microsoft Fabric solutions and highly automated data workflows using dbt, Flyway, Jenkins, and Python. Adept at integrating advanced AI/ML capabilities to drive business value, with hands-on experience in Generative AI (LLMs, RAG, Agentic Workflows) and predictive modeling with Snowpark ML. A proven leader of cross-functional teams who excels at translating complex business requirements into technical roadmaps and driving operational efficiency, and founder and leader of the Saint Louis Microsoft Fabric User Group.

SKILLS

- **Cloud Data Platforms:** Snowflake, Microsoft Fabric, AWS, Databricks, Vertica
- **Databases:** Redshift, Impala, PostgreSQL, Oracle, SQL Server, MongoDB, Azure Data Lake Gen2, AWS S3
- **AI & Machine Learning:** Generative AI (GPT-4, Gemini, Anthropic), Prompt Engineering, RAG Architectures, Agentic Workflows, MCP, LangChain, Vector Search (Pinecone, FAISS), Snowpark ML, MLflow, scikit-learn, Model Governance & Explainability
- **Infrastructure as Code & DevOps:** Terraform, Flyway, Jenkins, CI/CD, Azure DevOps, Docker, Git
- **Big Data:** Hadoop, Kafka, HDFS, Hive, Cassandra, MapReduce, Spark (PySpark, SparkSQL), ElasticSearch, Airflow
- **Programming:** SQL, Python (Pandas, NumPy, scikit-learn), T-SQL, DAX, KQL, PL/SQL, Bash
- **Reporting & BI:** Power BI, Tableau, Sigma, SAP Business Objects
- **Tools:** Azure Databricks, Azure Synapse Analytics, Azure Data Factory, Jira, ServiceNow, ODI 12
- **Data Engineering & Modeling:** Data warehouse design, ETL/ELT, data quality, Data Governance, Data Catalog, Dimensional modeling, Normalization, Schema Design

WORKING EXPERIENCE

Senior Snowflake Engineer

May 2026 – Present

Reinsurance Group of America · Full-time · Chesterfield, Missouri · Hybrid

- Serve as a Senior Snowflake Engineer within the enterprise Platform Team, responsible for administering and supporting the Snowflake data platform across multiple environments.
- Manage end-to-end Snowflake user access, including RBAC design, role hierarchy, grants, privileges, onboarding, offboarding, access reviews, and production permission support.
- Build and maintain Snowflake platform infrastructure, including warehouses, databases, schemas, roles, resource monitors, integrations, stages, network policies, and environment configurations.
- Automated Snowflake infrastructure provisioning using Terraform, enabling consistent deployment of Snowflake objects, access controls, warehouses, and platform configurations across Dev, QA, and Production.
- Integrated Jenkins-based CI/CD pipelines to support automated Snowflake deployments, infrastructure changes, schema updates, access changes, and release management activities.
- Support Snowflake DevOps operations by standardizing deployment workflows, managing environment configurations, and coordinating platform changes with data engineering and application teams.
- Monitor and optimize Snowflake performance by analyzing warehouse usage, query history, workload patterns, clustering needs, and compute resource utilization.
- Drive Snowflake FinOps practices by tracking credit consumption, tuning warehouse sizes, implementing auto-suspend/auto-resume standards, configuring resource monitors, and identifying cost optimization opportunities.
- Administer Snowflake security controls, including secure access policies, service accounts, authentication methods, data sharing permissions, masking policies, and audit monitoring.

Information Management Administrator

Sep 2024 – May 2026

neteffects · Contract · Chesterfield, Missouri · Hybrid

- Implemented and managed the Snowflake data platform, leveraging key features like Time Travel, Zero-Copy Cloning, and Secure Data Sharing to optimize data operations and governance.
- Developed and maintained ETL pipelines using dbt and Python, integrated with Jenkins for CI/CD, for improvement in data processing efficiency.
- Utilized Snowflake's AI and machine learning capabilities, including Snowpark for Python and integration with LLMs, to enhance data analytics and predictive modeling.
- Designed and implemented data models and workflows in Snowflake, ensuring scalability and performance across multiple data warehouses and compute clusters.
- Employed containerization techniques and automated workflows using Jenkins, improving efficiency and scalability.
- Applied data governance frameworks and tools such as Data Catalog, Data Lineage, and Data Profiling to ensure high-quality data.

- Provided technical support and guidance to data users, ensuring appropriate access and permissions, and collaborated with Enterprise Data Lakes for efficient data flows.

Information Technology Technical Associate
Southern Illinois University Edwardsville · Full-time · Edwardsville, Illinois · On-site

Feb 2024 – Oct 2024

- Implemented Microsoft Fabric from scratch, integrating Azure data services to support comprehensive data analytics projects.
- Applied data science techniques, including regression analysis and machine learning (RandomForest, RNN), to predict student retention and enrollment trends.
- Managed data extraction, transformation, and loading (ETL) using Fabric Notebooks and Azure Data Factory.
- Developed and maintained analytical dashboards and reports in Power BI, enhancing decision-making processes. Collaborated with university departments to understand their data needs, contributing to a 35% increase in data-driven decisions.

Data Analyst – Graduate Assistant
Southern Illinois University Edwardsville · Part-time · Edwardsville, Illinois

Jun 2022 – Dec 2023

- Led the design and development of visually appealing analytical dashboards and reports using Microsoft Power BI, Excel, and Access, which enhanced data visibility and resulted in a 20% increase in timely decision-making based on enrollment numbers and admission trends.
- Utilized SQL and data management techniques to extract, transform, and load data from university databases; improved data accuracy by 15% through data cleansing and transformation processes.
- Collaborated with stakeholders to define data requirements and specifications, built strong relationships, and fostered a data-driven culture, leading to a 30% increase in data-driven decisions.

Data Engineer – Analytics
Tata Consultancy Services · Full-time · Hyderabad, Telangana, India

Jun 2019 – Jan 2022

- Led dashboard development / analytics initiatives — including data prep by extracting data from different databases like AWS and SQL Server, using Python to build out predictive models and Tableau for developing the presentation layer in customer-facing dashboards to ensure data integrity and improve data quality, resulting in a 15% enhancement in data reliability.
- Conducted advanced statistical analyses, driving data-driven decision-making and contributing to a 12% increase in operational efficiency.
- Crafted compelling data visualizations using Tableau and Power BI, enhancing data-driven storytelling and achieving a 10% rise in stakeholder engagement.

CERTIFICATIONS

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| • SnowPro Core Certification | Dec 2024 – Dec 2026 |
| • Microsoft Certified: Fabric Analytics Engineer Associate | Mar 2024 – Mar 2026 |
| • Microsoft Certified: Power BI Data Analyst Associate | Feb 2024 – Feb 2026 |
| • Microsoft Certified: Azure AI Engineer Associate | May 2024 – May 2026 |
| • AWS Certified Cloud Practitioner | Sep 2023 – Sep 2026 |
| • SQL (Intermediate) Skills Certification | Dec 2023 |

EDUCATIONAL QUALIFICATIONS

Master of Science — Southern Illinois University Edwardsville
Major in Computer Management and Information Systems · GPA – 3.88 · Member, Entrepreneurship Club at SIUE

Jan 2022 – Dec 2023

Relevant coursework: Information Systems and Technology; Introduction to Business Intelligence and Analytics; Managing the Cloud; Enterprise Resource Planning (SAP); Project Management Fundamentals and Best Practices; Database Design; Software Systems Design; Quantitative Analysis; Project Procurement and Risk Management; Project Management Standard Processes.

Bachelor of Technology — Kakatiya Institute of Technology & Science, Warangal
Electrical and Electronics Engineering · GPA – 8.12 · EEE Association Event Manager; Sumshodini (technical event) Organiser; Cultural & Sports Teams

2015 – 2019

Relevant coursework: Programming in C; Object-Oriented Programming through C++; Data Structures; Entrepreneurship Development; AI Technologies in Electrical Engineering; Smart Electric Grid.

PROJECTS

- **KavachAI — Open-Source Privacy-Preserving AI Assistant:** Architected and developed an open-source full-stack, production-ready AI assistant focused on data privacy and security. Engineered a hybrid AI system that sanitizes sensitive information using local language models (via Ollama) before sending data to cloud-based LLMs (OpenAI, Google) for complex processing, ensuring confidential data never leaves the user's local infrastructure and addressing compliance needs like GDPR and HIPAA. Built with FastAPI, Next.js + TypeScript, and fully containerized with Docker Compose.
- **Snowflake Data Management & Optimization Platform:** Developed a full-stack platform providing intelligent insights for Snowflake environments — an AI-powered query advisor using the OpenAI API, real-time performance monitoring dashboards, data lineage tracking, and automated ETL pipeline management. Built with React + TypeScript, a Node.js/Express backend, PostgreSQL, and direct integration with the Snowflake SDK.

- **Snowflake Cortex Agent Integration via MCP Server:** Built and configured a Model Connectivity Protocol (MCP) server to connect Snowflake Cortex Agents with the Cursor AI application using Python, the MCP SDK, and Snowflake's Cortex Search service — engineering the end-to-end integration from environment setup and secure token generation to server deployment and client configuration. [\[github.com/AkhilGurrapu/snowflake-mcp-cortex-agent\]](https://github.com/AkhilGurrapu/snowflake-mcp-cortex-agent)
- **GitSarva — Interactive Git Learning Platform:** Developed a browser-first progressive web app to teach Git and GitHub through hands-on tutorials, featuring a custom browser-based Git simulation engine with a live terminal and real-time visual Git tree. Built with React + TypeScript, an Express.js server, and a Supabase (PostgreSQL) backend managed by Drizzle ORM. [\[github.com/AkhilGurrapu/gitsarva\]](https://github.com/AkhilGurrapu/gitsarva)
- **FinanceAI (Microsoft Fabric AI Hackathon):** A Power BI project for real-time stock analysis of the top-20 market stocks using Microsoft Fabric, the OpenAI API, and a RandomForestRegressor model for predictive insights — integrating company profiles, financial news, and stock predictions in a dedicated Lakehouse workspace. [\[github.com/AkhilGurrapu/FabricAIHack-StockReport\]](https://github.com/AkhilGurrapu/FabricAIHack-StockReport)
- **PowerLens — Power BI & Fabric License Calculator:** A responsive web application that helps organizations determine the most cost-effective Microsoft Power BI and Fabric licensing strategy via a multi-step wizard analyzing user count, data volume, and premium feature requirements. Built with HTML, CSS, and JavaScript. [\[github.com/AkhilGurrapu/powerlens\]](https://github.com/AkhilGurrapu/powerlens)
- **Stock Beta Calculator using CAPM:** A Python/Streamlit application using pandas, NumPy, and yfinance to calculate a stock's beta and expected returns using the Capital Asset Pricing Model based on user inputs. [\[capitalassestpricingmodel.streamlit.app\]](https://capitalassestpricingmodel.streamlit.app)

ACHIEVEMENTS

- TCS Employee of the Year — 2021.
- First international student to earn a full-time position at SIUE.
- Leader of the Saint Louis Microsoft Fabric / Power BI User Group, actively fostering the data community.
- First Graduate Assistant at SIUE to receive a salary increase.